STELLAR NUCLEOSYNTHESIS — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: April 15, 2026

Diagram-based questions on Stellar Nucleosynthesis

Question 1

10.0 points

Examine the proton-proton (p-p) chain reaction depicted in the diagram below for main sequence stars. Identify and explain the sequence of nuclear reactions occurring in the p-p chain, including the role of neutrinos and gamma rays in energy transport within the star.

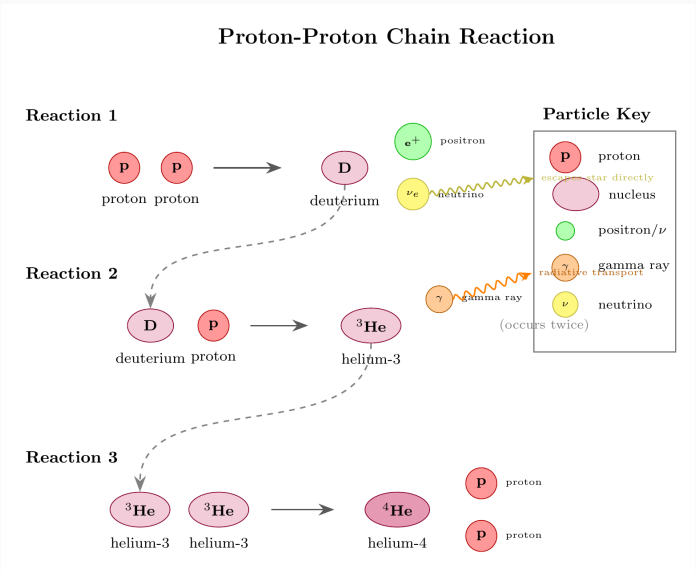

Fig. 1. A diagram of the proton-proton (p-p) chain reaction in main sequence stars. The diagram should include the following stages: 1. Two protons (hydrogen nuclei) combine to form a deuterium nucleus, a positron, and a neutrino. 2. The deuterium nucleus fuses with another proton to form a helium-3 nucleus and a gamma ray. 3. Two helium-3 nuclei combine to form a helium-4 nucleus, releasing two protons. Include labels for each particle and reaction, with arrows indicating the sequence of reactions. Highlight the emission of neutrinos and gamma rays as part of the energy transport process within the star. IMPORTANT correction needed: The diagram explicitly labels every reactant and product in. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram incorrectly shows a positron and. Regenerate using EXACTLY the values and sequences specified.

Question 2 10.0 points

Examine the CNO cycle reaction network in the diagram below for massive stars. Describe how this cycle contributes to the stellar nucleosynthesis process and compare its efficiency to the proton-proton chain in terms of energy production.

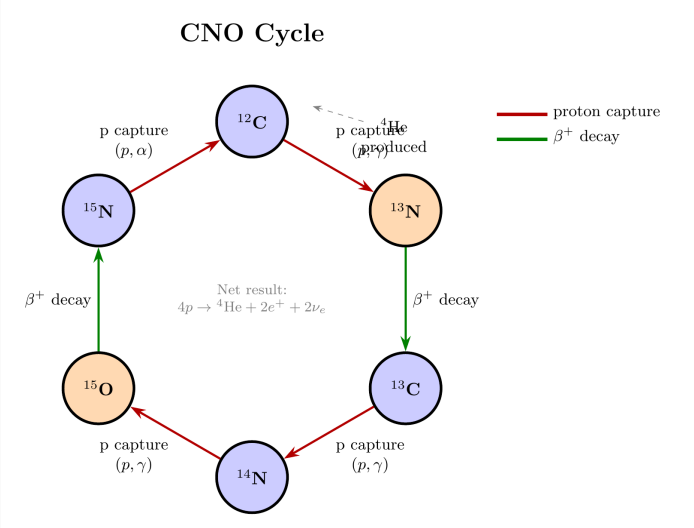

Fig. 2. Diagram of the CNO cycle reaction network for massive stars. Include the main nuclei involved: carbon-12 (C-12), nitrogen-14 (N-14), oxygen-16 (O-16), and the reactions that convert these nuclei through the cycle. Show the sequence of reactions starting with C-12 capturing a proton to form N-13, followed by beta decay to C-13, proton capture to N-14, proton capture to O-15, beta decay to N-15, and finally proton capture to return to C-12. Label each nucleus and reaction step clearly, and indicate energy release at each step where applicable. Use arrows to show the direction of the cycle and highlight the cyclical nature of the process. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram shows the wrong isotope of Oxygen; it. Regenerate using EXACTLY the values and sequences specified. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram shows the final step of the cycle as. Regenerate using EXACTLY the values and sequences specified.

Question 3

10.0 points

Interpret the Hertzsprung-Russell (H-R) diagram with evolutionary tracks in the diagram below. Explain how this diagram represents the life cycles of different mass stars and identify the key stages for a one-solar-mass star.

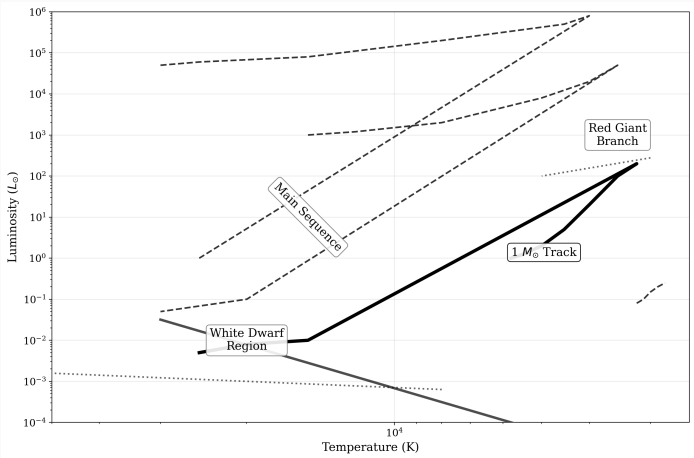

Fig. 3. Hertzsprung-Russell (H-R) diagram showing evolutionary tracks for stars of different masses. The diagram should include axes for luminosity (vertical) and temperature (horizontal, decreasing to the right). Key stages such as the main sequence, red giant, and white dwarf phases should be labeled, with a highlighted track for a one-solar-mass star showing its progression through these stages. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram shows temperature increasing to the right,. Regenerate using EXACTLY the values and sequences specified. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram shows temperature increasing to the. Regenerate using EXACTLY the values and sequences specified.

Question 4 10.0 points

Examine the binding energy per nucleon curve in the diagram below and the iron peak. Discuss how this curve elucidates the energy production limits in stellar fusion and why iron signifies an endpoint for nucleosynthesis in stars.

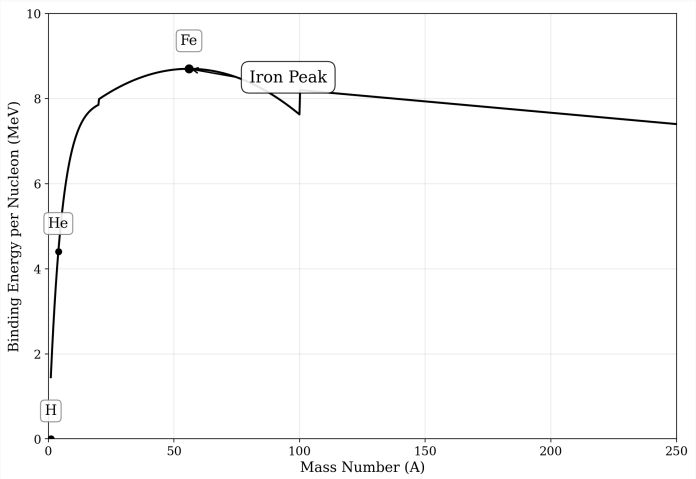

Fig. 4. Binding energy per nucleon curve showing the trend from light to heavy elements, highlighting the iron peak. The x-axis represents the mass number, and the y-axis represents the binding energy per nucleon. The curve should show a peak around iron (Fe), illustrating why iron is the endpoint for nucleosynthesis in stars.

Question 5 10.0 points

Examine the onion-shell model diagram of a pre-supernova massive star's internal structure shown below. Describe the different shells and their associated nuclear processes, explaining how this structure leads to a supernova event.

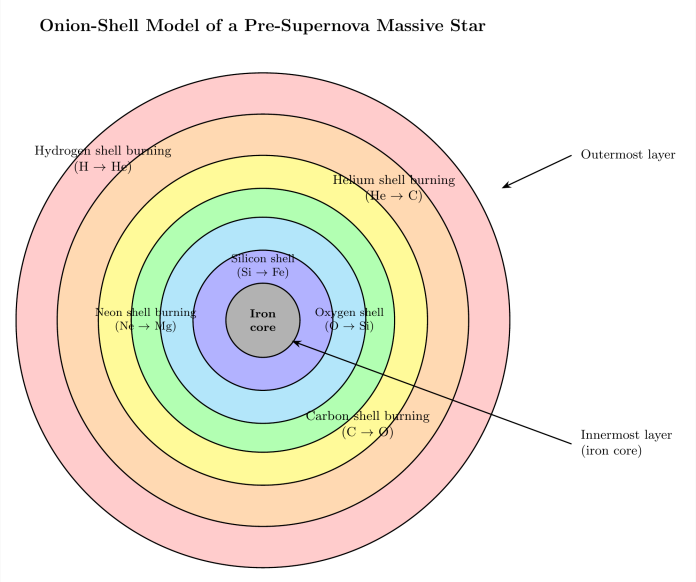

Fig. 5. A diagram of the onion-shell model of a pre-supernova massive star, showing concentric layers from the outermost to the innermost. Each layer should be labeled with the element being fused and the nuclear process occurring. From outside to center: Hydrogen shell burning (H to He), Helium shell burning (He to C), Carbon shell burning (C to O), Oxygen shell burning (O to Ne), Neon shell burning (Ne to Mg), Magnesium shell burning (Mg to Si), Silicon shell burning (Si to Fe core). The diagram should clearly show the progression towards the iron core, which is the final stage before a supernova event. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram is missing the "Mag. Regenerate using EXACTLY the values and sequences specified. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram shows an incorrect sequence of nuclear fusion. Regenerate using EXACTLY the values and sequences specified.